

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	E.B.ASHWINI
Register Number	192511422

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)

Total Marks: 40

Question Count: 4

Duration : 60 Minutes

Code of Conduct

I _____E.B.ASHWINI/192511422_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ashwini_____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application

Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

EXPERIMENT 1: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR LIBRARY BOOK RESERVATION SYSTEM FOR SIMATS LIBRARY USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS INCLUDE THE NECESSARY FIELDS SUCH AS TITLE OF THE BOOK, AUTHOR, PUBLICATIONS, YEAR, EDITION, NO OF COPIES, RACK NO, STATUS (TAKEN/RETURNED) ETC.

AIM

To create a simple cloud-based Library Book Reservation System using a Software as a Service (SaaS) platform to manage library books and reservations.

PROCEDURE

Step 1: Open a browser and log in to a SaaS platform such as Zoho Creator.

Step 2: Create a new application named Library Book Reservation System.

Step 3: Create a form and add fields such as Book Title, Author, Publication, Year, Edition, No. of Copies, Rack No., and Status.

Step 4: Set suitable data types for each field and save the form.

Step 5: Enter sample book details into the application.

Step 6: Search for a required book and check its availability.

Step 7: Update the status as Taken when a book is reserved and Returned when it is returned.

Step 8: Verify that the records are stored and updated successfully in the cloud.

IMPLEMENTATION

Library Book Reservation

Dashboard

Reservations

+ Reservations

Reservations Report

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Books

Dashboard

Divyasree

Reservations

Member

Book

Status

Reservation Date

Due Date

Return Date

Reservation Type

Notes

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Reset

Library Book Reservation

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Divyasree

Reservations Report

Member	Book	Status	Reservation Date	Due Date	Return Date	Reservation Type	Notes
M001 M002 M003 M004 M005	Harry Potter	Returned	19-Jul-2026	26-Jul-2026	24-Jul-2026	Priority	Renewed reservation for summer reading
M001 M002 M003	Harry Potter	Returned	18-Jul-2026	25-Jul-2026	23-Jul-2026	Standard	Book damaged during transport
M001 M002 M004 M005	Harry Potter 1984 To Kill a Mockingbird	Taken	17-Jul-2026	24-Jul-2026	22-Jul-2026	Priority	Late return fee applies here
M001	Harry Potter 1984	Returned	16-Jul-2026	23-Jul-2026	21-Jul-2026	Standard	Extended return due to rain
M001 M002 M003	Harry Potter To Kill a Mockingbird Pride and Prejudice	Returned	15-Jul-2026	22-Jul-2026	20-Jul-2026	Priority	Book returned early today

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Library Book Reservation

Trial expires in 4 daysUpgradeEdit this applicationHelp

Library Book Reservation

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Members

Member Name

Register Number

Phone

+91

81234 56789

Registration Date

dd-MM-yyyy

Status

Active

Inactive

Other

Membership Type

-Select-

Last Updated

dd-MM-yyyy HH:mm:ss

Submit

Reset

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Members Report

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+

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Member Name	Register Number	Phone	Member Id	Registration Date	Status	Membership Type	Last Updated
Emma White	M005	+18093507787	5	12-May-2024	Active	Student	31-Jul-2026 16:00:00
David Lee	M004	+17792102257	4	05-Apr-2024	Active	Staff	31-Jul-2026 14:20:00
Carol Martinez	M003	+18723587332	3	10-Mar-2024	Inactive	Visitor	31-Jul-2026 09:15:00
Bob Williams	M002	+19857253938	2	20-Feb-2024	Inactive	Visitor	31-Jul-2026 11:45:00
Alice Johnson	M001	+17349721474	1	15-Jan-2024	Inactive	Visitor	31-Jul-2026 10:30:00

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Library Book Reservation

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Books

Book Name

Title

Author

Publication

Year

Edition

Number Of Copies

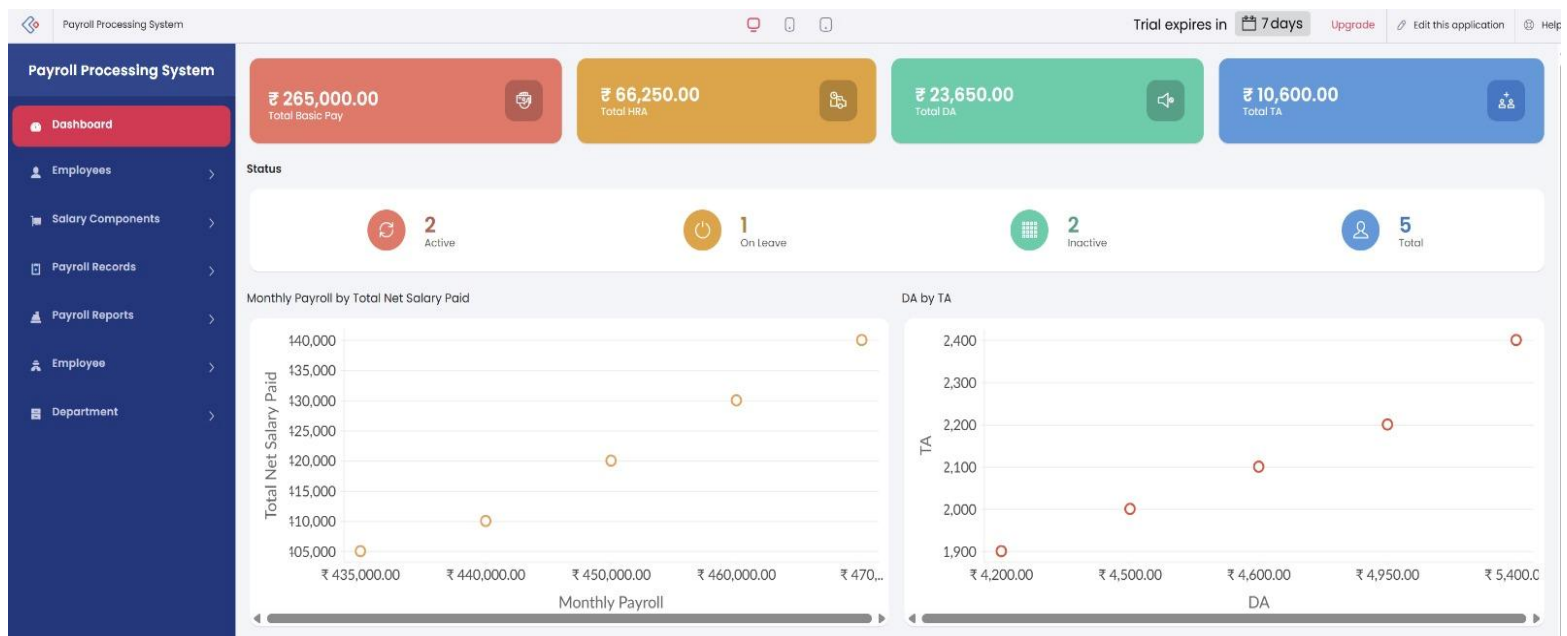
Rack Number

Total Copies

Available Copies

Submit

Reset



RESULT

A cloud-based Library Book Reservation System was successfully created using a SaaS platform. The system stores book details, manages reservations, updates book status, and provides secure cloud-based access to library records.

EXPERIMENT 2: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR PAYROLL PROCESSING SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS INCLUDE THE NECESSARY FIELDS SUCH AS NAME OF THE EMPLOYEE, EXPERIENCE IN PRESENT ORG , OVER ALL EXPERIENCE, BASIC PAY, HRA, DA, TA, NET PAY, GROSS PAY ETC.

AIM

To create a simple cloud-based **Payroll Processing System** using a **Software as a Service (SaaS)** platform to manage employee payroll details.

PROCEDURE

Step 1: Open a browser and log in to a SaaS platform such as Zoho Creator.

Step 2: Create a new application named Payroll Processing System.

Step 3: Create a form and add fields such as Employee Name, Experience in Present Organization, Overall Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.

Step 4: Set appropriate data types for the fields and save the form.

Step 5: Enter sample employee details and salary information.

Step 6: Calculate and enter the Gross Pay and Net Pay.

Step 7: Add and update employee payroll records as required.

Step 8: Verify that all payroll records are stored and accessible through the cloud.

IMPLEMENTATION



Form



Report



Page



Workflow

How would you like to create your form?

On my own



From scratch



Import with data



Using Zia

Create from Scratch

Name form

Examples: Add Order, Staff Details

Create Form

Payroll Processing System

Trial expires in 4 daysUpgradeEdit this applicationHelp

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Employee Name

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Overall Experience

Gender

Hire Date

Contact Info

Manager

Status

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Employees Report

Employee Name

Department

Overall Experience

Gender

Hire Date

Contact Info

Manager

Status

	Lisa Anderson	Finance and Accounting	4	Male	05-Apr-2022	+17725441711	David Lee	On Leave
	Robert Taylor	Information Technology	12	Female	10-Jan-2015	+18194533394	James Wilson	Active
	Emily Rodriguez	Finance and Accounting	3	Female	20-Sep-2023	+18503082804	David Lee	Active
	Michael Chen	Marketing and Sales	8	Male	02-Jun-2018	+19563023036	James Wilson	Inactive
	Sarah Connor	Finance and Accounting	5	Female	15-Mar-2021	+18406267877	James Wilson	Inactive

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Payroll Processing System

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Salary Components

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Salary Components

Employee

-Select-

Basic Pay

#####.##

HRA

#####.##

DA

#####.##

TA

#####.##

Effective Date

dd-MMM-yyyy

Status

Active

Inactive

Other

Submit

Reset

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Salary Components Report

Employee

Basic Pay

HRA

DA

TA

Effective Date

Status

Carol	₹ 52,000.00	₹ 13,000.00	₹ 4,600.00	₹ 2,100.00	25-Jul-2026	Active
Robert	₹ 48,000.00	₹ 12,000.00	₹ 4,200.00	₹ 1,900.00	10-Jul-2026	Inactive
Robert	₹ 60,000.00	₹ 15,000.00	₹ 5,400.00	₹ 2,400.00	01-Aug-2026	Active
Robert	₹ 55,000.00	₹ 13,750.00	₹ 4,950.00	₹ 2,200.00	15-Jul-2026	Active
Carol	₹ 50,000.00	₹ 12,500.00	₹ 4,500.00	₹ 2,000.00	01-Jul-2026	Active

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Show Summary

Payroll Processing System

4 days

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Payroll Records

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Payroll Records

Employee

-Select-

Pay Period

Gross Pay

#####.##

Net Pay

#####.##

Deductions

#####.##

Payment Date

dd-MM-yyyy

Processed By

Status

-Select-

Submit

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Payroll Records Report

Employee

Pay Period

Gross Pay

Net Pay

Deductions

Payment Date

Processed By

Status

Robert	Jul-2026	₹ 5,900.00	₹ 5,150.00	₹ 750.00	31-Jul-2026	Eva Brown	Failed
Elena	Jul-2026	₹ 7,100.00	₹ 6,300.00	₹ 800.00	31-Jul-2026	David Lee	Failed
David	Jul-2026	₹ 4,800.00	₹ 4,200.00	₹ 600.00	31-Jul-2026	Carol White	Failed
Carol	Jul-2026	₹ 6,200.00	₹ 5,400.00	₹ 800.00	31-Jul-2026	Bob Jones	Failed
David	Jul-2026	₹ 5,500.00	₹ 4,850.00	₹ 650.00	31-Jul-2026	Alice Smith	Failed

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Show Summary

Payroll Processing System

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Employee

Display Name

Employee ID No.

First Name *

Gender

Last Name *

Middle Name

Print On Check as

Suffix

Title

Submit

Reset

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Edit this application

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Employee Report

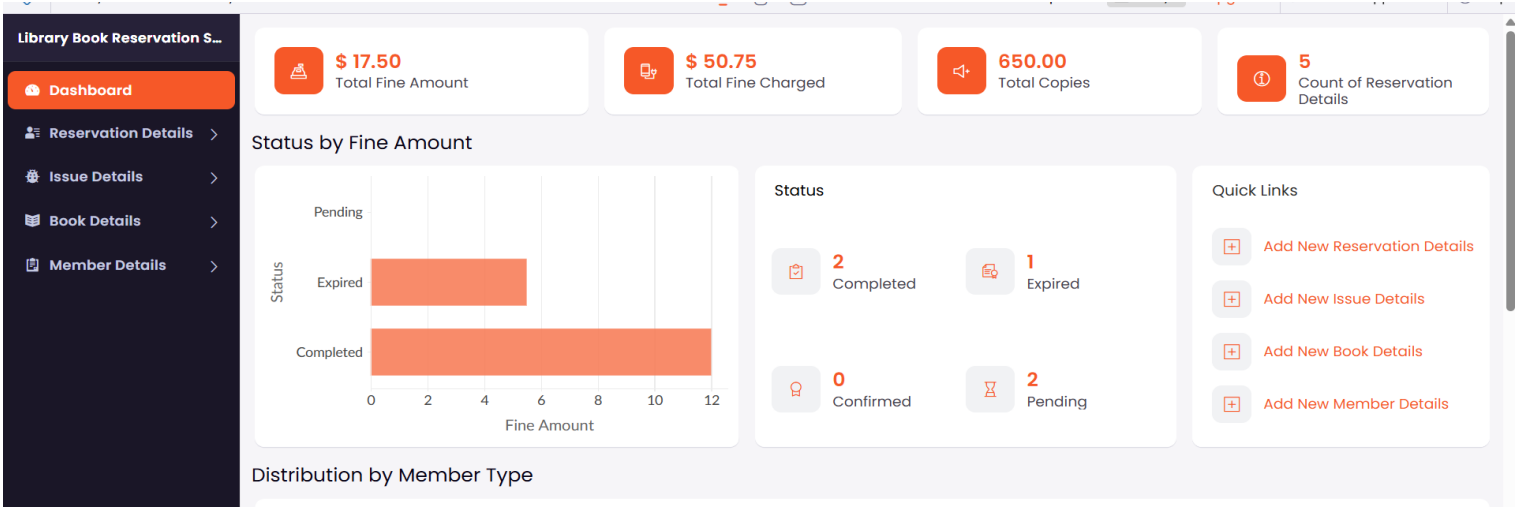
Search

+ Employee

...

	Display Name	Employee ID No.	First Name	Gender	Last Name	Middle Name	Print On Check as	Suffix	Title
	Eva Garcia	EMP005	Elena	Female	Garcia	Maria	E	Sr	Consultant
	David Lee	EMP004	David	Male	Lee	Michael	D	Jr	Analyst
	Carol White	EMP003	Carol	Female	White	Ann	C	III	VP
	Bob Smith	EMP002	Robert	Male	Smith	James	B	Sr	Director
	Alice Johnson	EMP001	Alice	Female	Johnson	Marie	A	Jr	Senior Manager

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Library Book Reservation S...

Dashboard

Reservation Details >

Issue Details >

Book Details >

Member Details >

Reservation Details Report

Book	Member	Reservation Date	Due Date	Status	Fine Amount	Reservation Notes
The Great Gatsby To Kill a Mockingbird 1984 Pride and Prejudice The Catcher in the Rye	Alice Johnson Bob Williams Carol Martinez David Lee Eva Chen	19-Jul-2026	26-Jul-2026	Pending	\$ 0.00	Summer reading club pickup
The Great Gatsby To Kill a Mockingbird 1984 Pride and Prejudice The Catcher in the Rye	Bob Williams Carol Martinez David Lee	18-Jul-2026	25-Jul-2026	Completed	\$ 12.00	Multiple overdue books fined
The Great Gatsby To Kill a Mockingbird 1984 Pride and Prejudice The Catcher in the Rye	Alice Johnson	17-Jul-2026	24-Jul-2026	Pending	\$ 0.00	New book reservation confirmed
The Great Gatsby To Kill a Mockingbird Pride and Prejudice	Alice Johnson Bob Williams	16-Jul-2026	23-Jul-2026	Expired	\$ 5.50	Late fee applied for overdue
To Kill a Mockingbird 1984	Alice Johnson Bob Williams Carol Martinez David Lee	15-Jul-2026	22-Jul-2026	Completed	\$ 0.00	Book returned early today

RESULT

A cloud-based Payroll Processing System was successfully created using a SaaS platform. The application stores employee payroll information, calculates salary details, updates records, and provides secure cloud-based access to payroll data.

EXPERIMENT 3: DEMONSTRATE VIRTUALIZATION BY INSTALLING TYPE-2 HYPERVISOR IN YOUR DEVICE, CREATE AND CONFIGURE VM IMAGE WITH A HOST OPERATING SYSTEM (EITHER WINDOWS/LINUX). CREATE A VIRTUAL MACHINE WITH 1 CPU, 2GB RAM AND 15GB STORAGE DISK USING A TYPE 2 VIRTUALIZATION SOFTWARE.

AIM

To install a Type-2 Hypervisor (VMware Workstation) and create a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB virtual hard disk using a Windows/Linux host operating system.

PROCEDURE

Step 1: Install and open VMware Workstation on the host computer.

Step 2: Click Create a New Virtual Machine and select Typical.

Step 3: Select the Windows/Linux ISO file for the guest operating system.

Step 4: Enter the virtual machine name and select the storage location.

Step 5: Set the virtual hard disk size to 15 GB.

Step 6: Click Customize Hardware and set Memory to 2 GB (2048 MB).

Step 7: Set the Processor to 1 CPU and 1 Core.

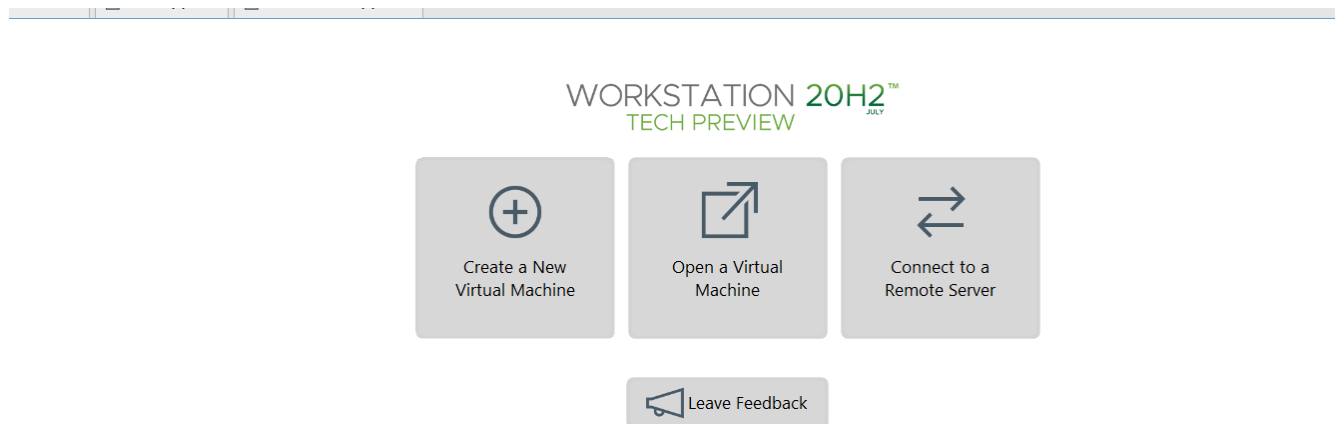
Step 8: Save the hardware settings and click Finish.

Step 9: Power on the virtual machine and install the guest operating system.

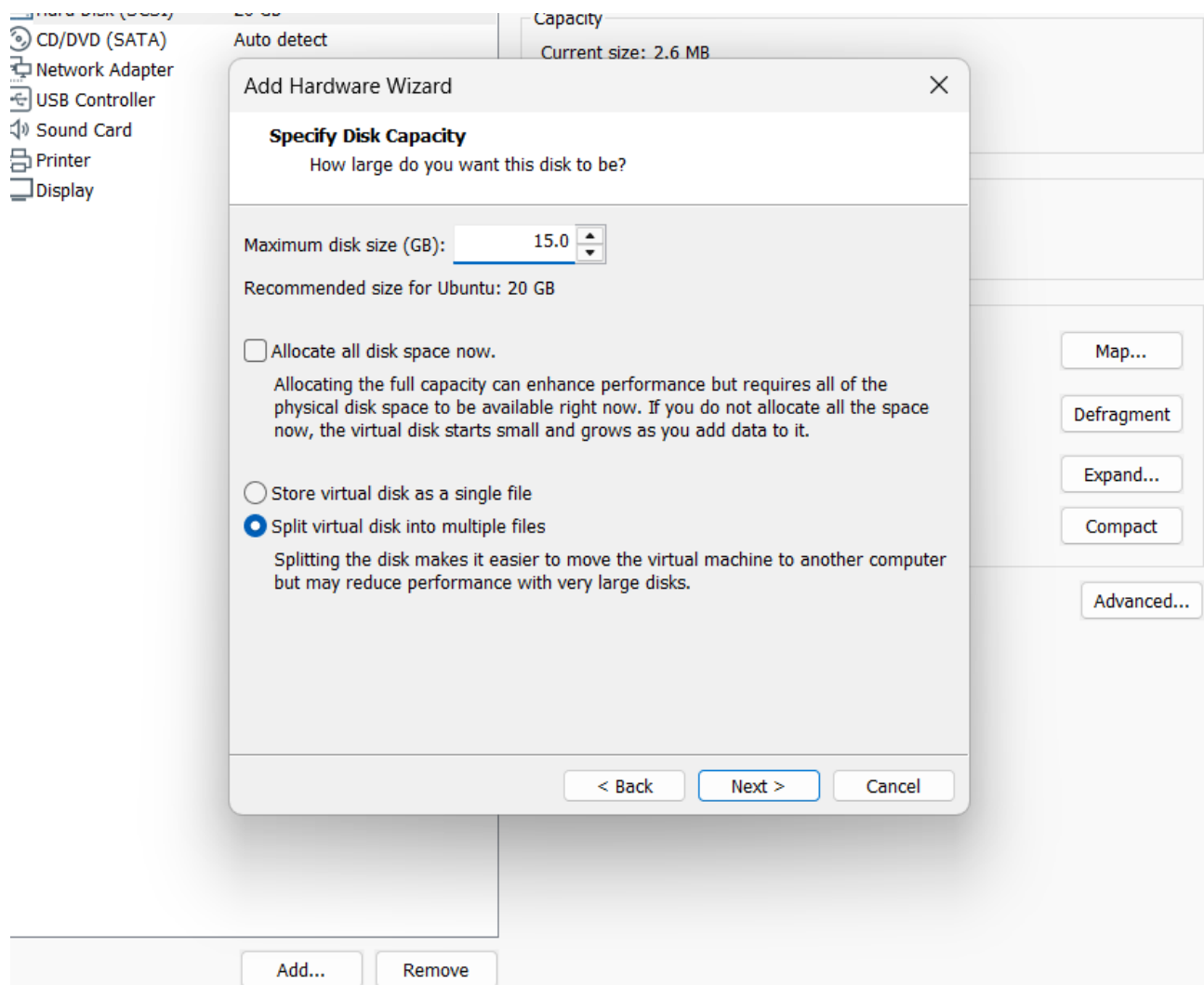
Step 10: Verify that the VM starts successfully with the configured CPU, RAM, and storage.

IMPLEMENTATION.

Step 1: Click Create a New Virtual Machine and select Typical



Step 2: Set the virtual hard disk size to 15 GB.



Step 3: Click Customize Hardware and set Memory to 2 GB (2048 MB).

The screenshot shows the 'Memory' configuration window for a virtual machine. On the left, a list of devices includes Memory (2 GB), Processors (2), New CD/DVD (SATA) (Auto detect), Network Adapter (NAT), USB Controller (Present), Sound Card (Auto detect), Printer (Present), and Display (Auto detect). The 'Memory' device is selected. The main area shows the memory allocation settings. A text box indicates 'Memory for this virtual machine: 2048 MB'. Below this is a slider with a scale from 512 MB to 128 GB. The slider is positioned at 2 GB. To the right of the slider, there are two horizontal lines: a blue line for 'Maximum recommended memory' at 13.2 GB and a green line for 'Recommended memory' at 2 GB. A legend on the right explains these lines: 'Maximum recommended memory (Memory swapping may occur beyond this size.)' and 'Recommended memory'.

Device	Summary
Memory	2 GB
Processors	2
New CD/DVD (SATA)	Auto detect
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

Memory

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: 2048 MB

128 GB -
64 GB -
32 GB -
16 GB -
8 GB -
4 GB -
2 GB -
1 GB -
512 MB -

Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.2 GB

Recommended memory

Step 4: Set the Processor to 1 CPU and 1 Core.

The screenshot shows the 'Processors' configuration window for a virtual machine. On the left, a list of devices includes Memory (2 GB), Processors (1), New CD/DVD (SATA) (Auto detect), Network Adapter (NAT), USB Controller (Present), Sound Card (Auto detect), Printer (Present), and Display (Auto detect). The 'Processors' device is selected. The main area shows the processor configuration. There are three dropdown menus: 'Number of processors:' set to 1, 'Number of cores per processor:' set to 1, and 'Total processor cores:' set to 1. Below these is the 'Virtualization engine' section with three checkboxes: 'Virtualize Intel VT-x/EPT or AMD-V/RVI' (checked), 'Virtualize CPU performance counters' (unchecked), and 'Virtualize IOMMU (IO memory management unit)' (unchecked).

Device	Summary
Memory	2 GB
Processors	1
New CD/DVD (SATA)	Auto detect
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

Processors

Number of processors: 1

Number of cores per processor: 1

Total processor cores: 1

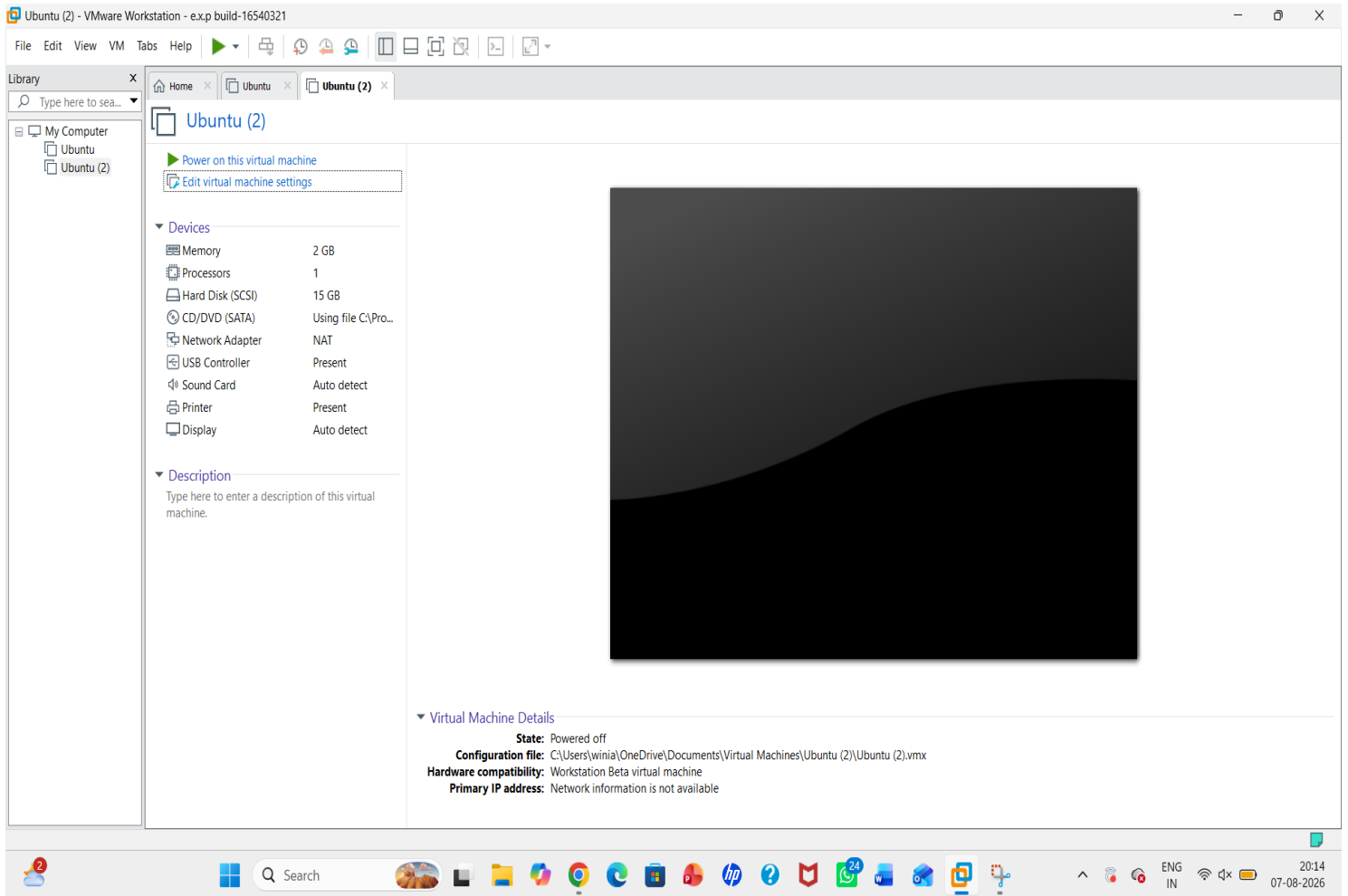
Virtualization engine

☒ Virtualize Intel VT-x/EPT or AMD-V/RVI

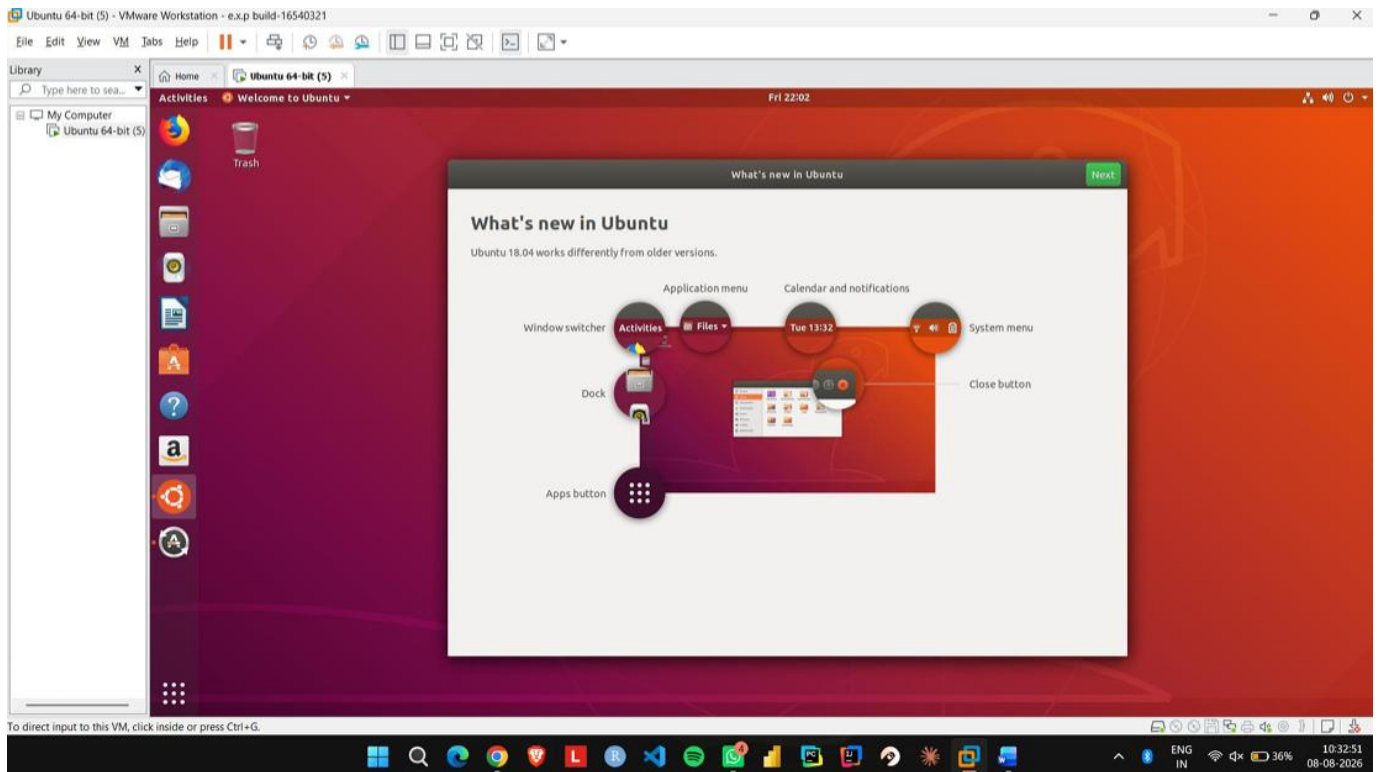
☐ Virtualize CPU performance counters

☐ Virtualize IOMMU (IO memory management unit)

Step 5: Power on the virtual machine and install the guest operating system



Step 6: Verify that the VM starts successfully with the configured CPU, RAM, and storage.



RESULT

A Virtual Machine was successfully created and configured using VMware Workstation with 1 CPU, 2 GB RAM, and a 15 GB virtual hard disk. The guest operating system was installed and operated successfully.

EXPERIMENT 4: DEMONSTRATE VIRTUALIZATION BY INSTALLING TYPE-2 HYPERVISOR IN YOUR DEVICE, CREATE AND CONFIGURE VM IMAGE WITH A HOST OPERATING SYSTEM (EITHER WINDOWS/LINUX). CREATE A CLONING OF A VM AND TEST IT BY LOADING THE PREVIOUS VERSION/CLONED VM

AIM

To create a clone of an existing Virtual Machine using VMware Workstation and verify its functionality by booting the cloned virtual machine.

PROCEDURE

Step 1: Open VMware Workstation and select the existing Virtual Machine.

Step 2: Shut down the Virtual Machine completely.

Step 3: Right-click the VM and select Manage → Clone.

Step 4: In the Clone Wizard, select Current State in the Virtual Machine.

Step 5: Select Full Clone and click Next.

Step 6: Enter a name and storage location for the cloned VM.

Step 7: Click Finish and wait until the cloning process is completed.

Step 8: Select the newly created cloned VM from the VMware library.

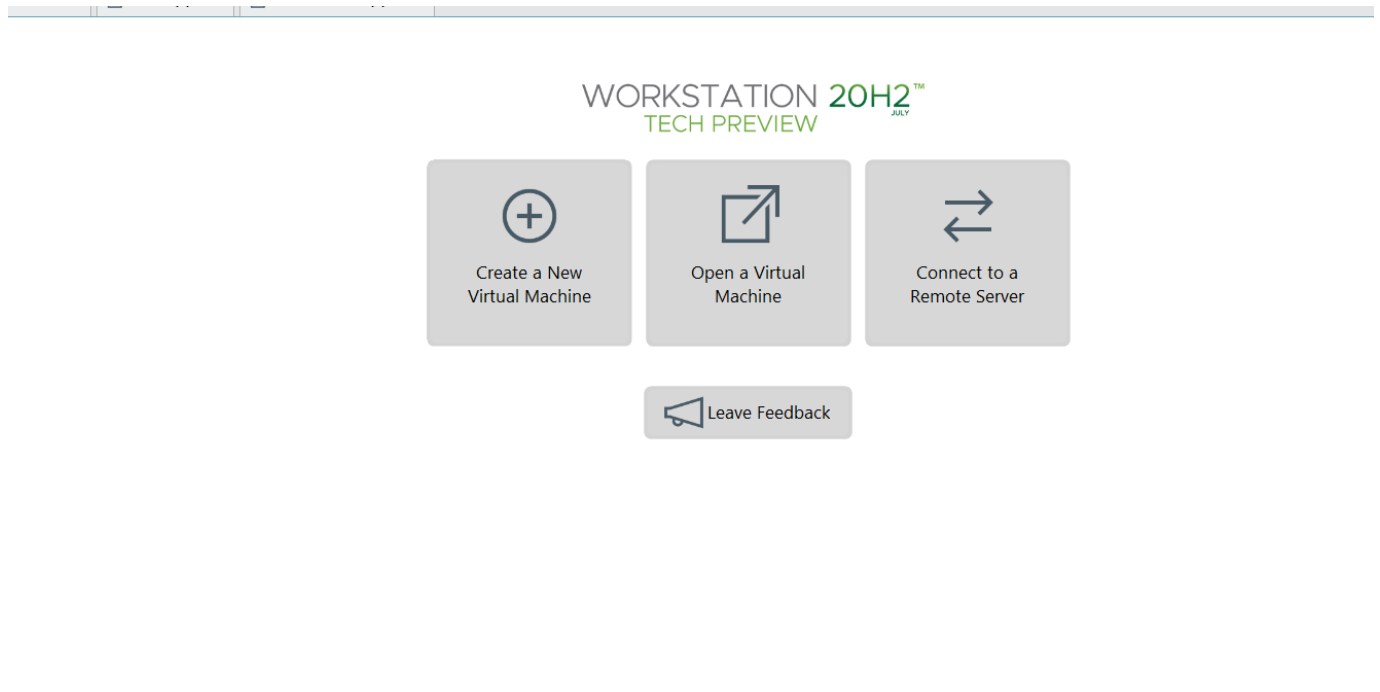
Step 9: Power on the cloned VM and wait for the operating system to load.

Step 10: Verify that the files, applications, and operating system work correctly.

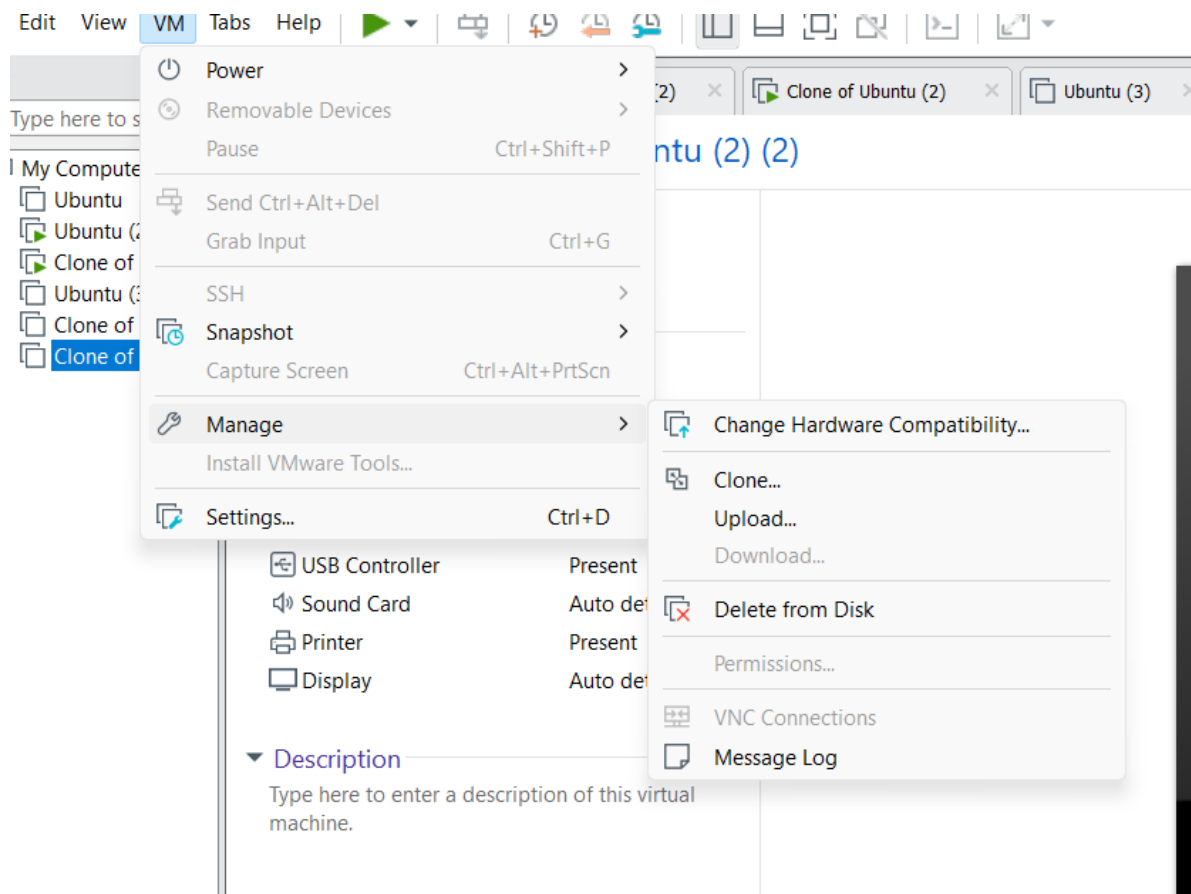
Step 11: Confirm that the cloned VM works independently from the original VM.

IMPLEMENTATION

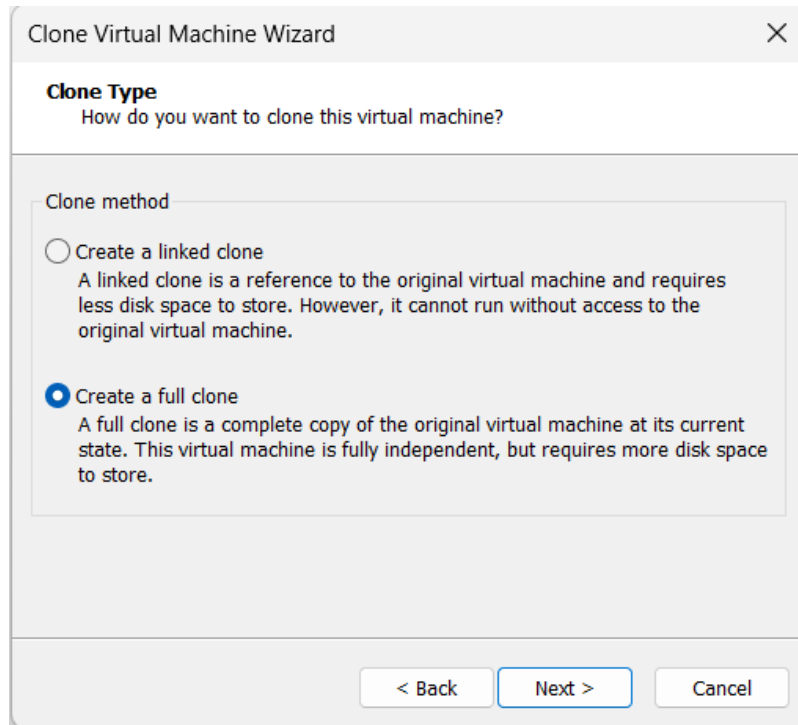
Step 1: Open VMware Workstation and select the existing Virtual Machine



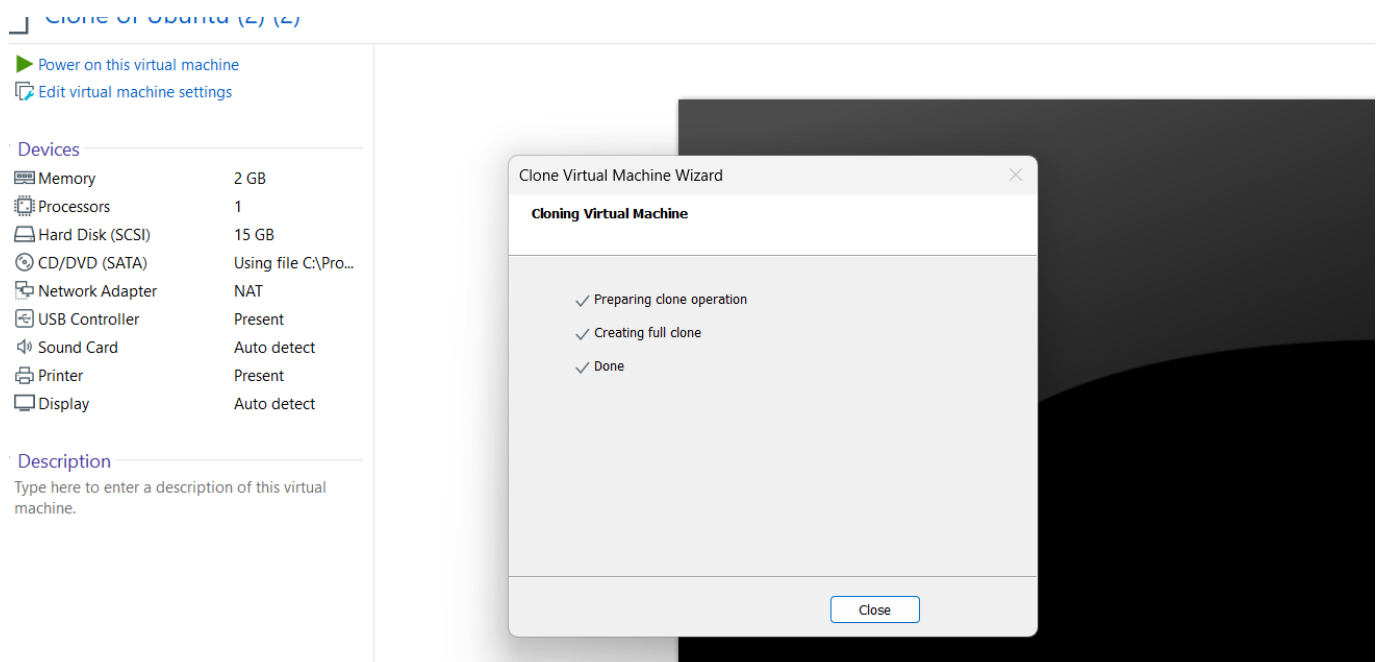
Step 3: Right-click the VM and select Manage → Clone.



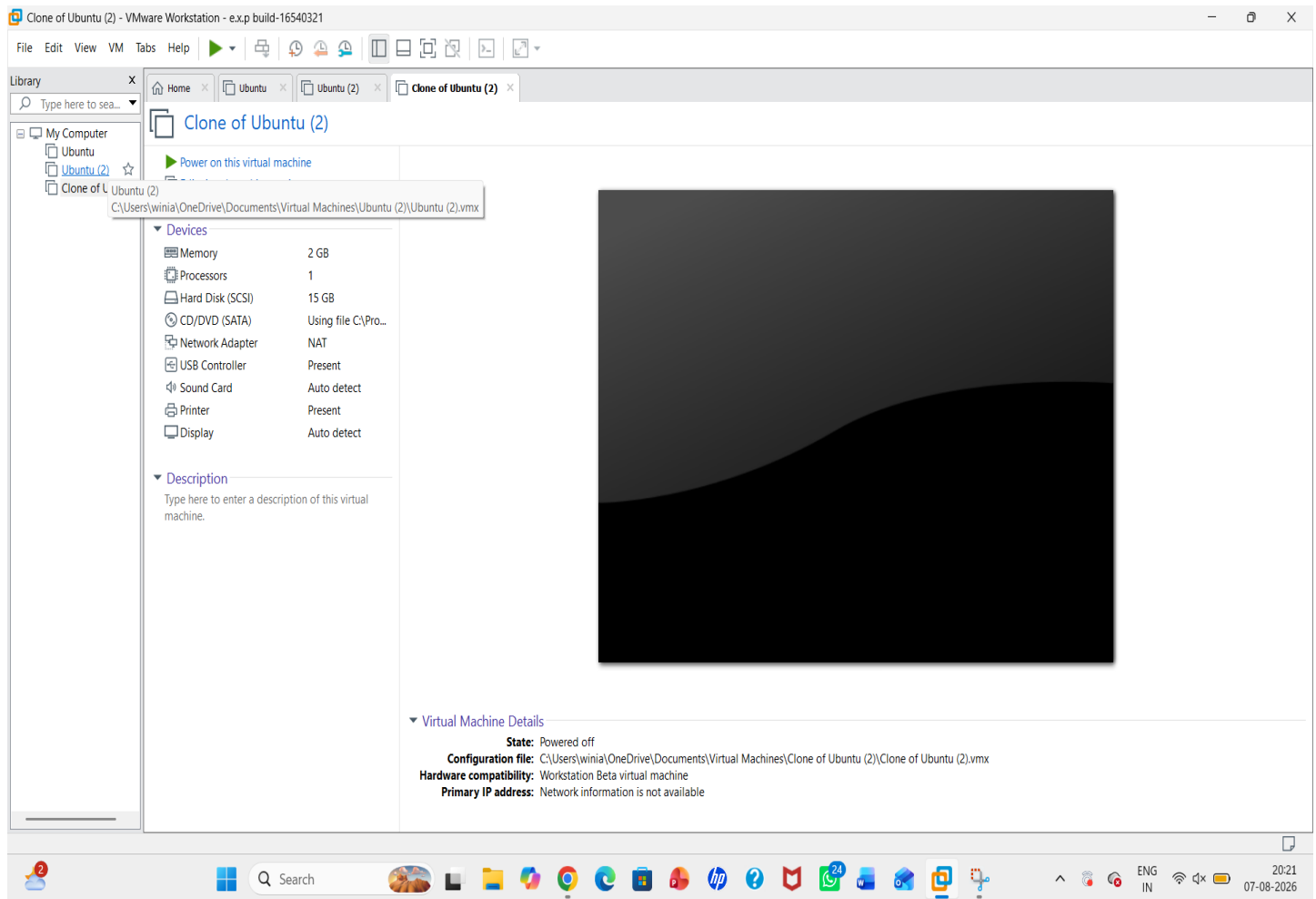
Step 4: Select Full Clone and click Next.



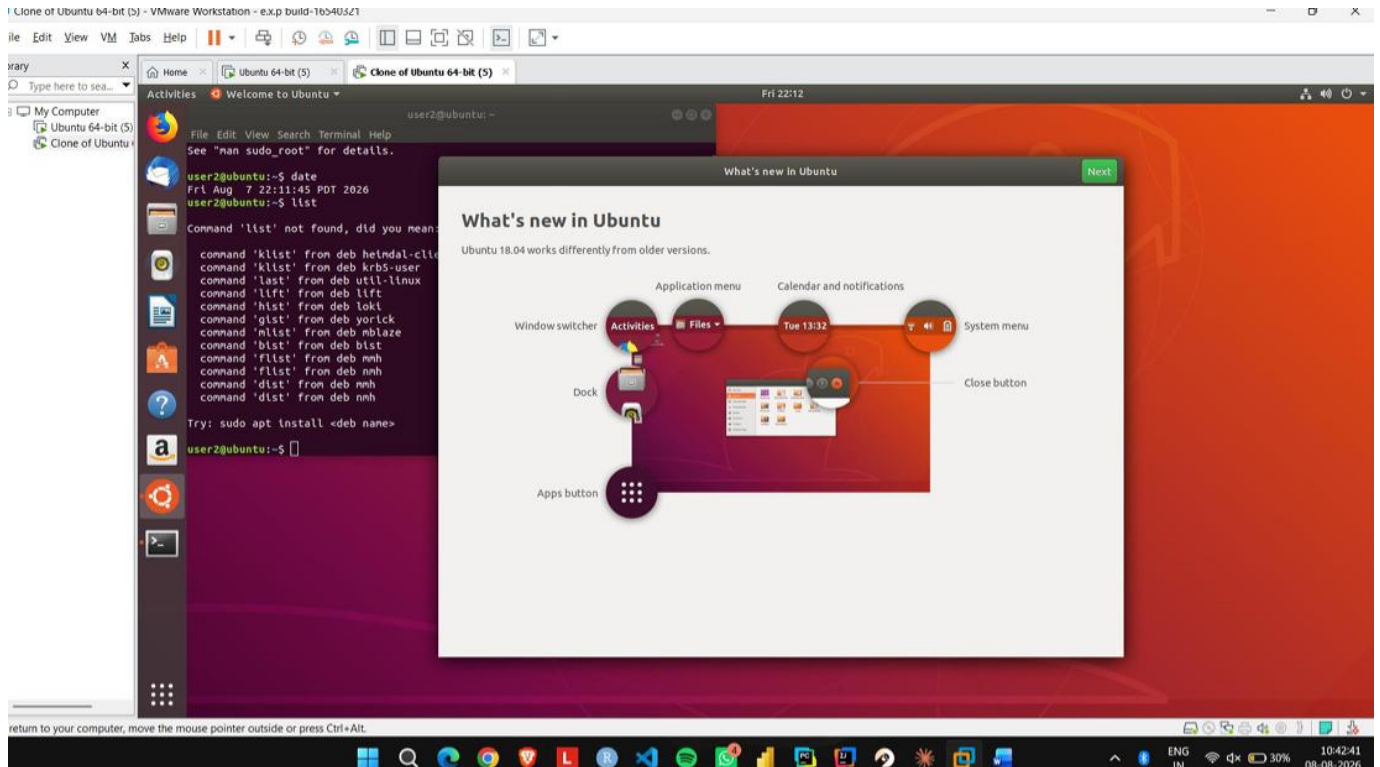
Step 5: Click Finish and wait until the cloning process is completed.



Step 5: Verify that the files, applications, and operating system work correctly.



Step 6: Confirm that the cloned VM works independently from the original VM.



RESULT

The virtual machine was successfully cloned using VMware Workstation. The cloned virtual machine booted correctly and functioned independently, demonstrating successful virtual machine cloning and virtualization.